Ruiwen WANG

Research Engineer — ML Systems & AI Infrastructure

Distributed LLM training · Parallelism & performance optimisation

Contact

Paris, France
wangrw0124@gmail.com
(+33) 6 63 11 34 66
French (EU) · open to relocation

Highlights

Production training
Scale: 16,384 accelerators
TeleChat3-MoE contributor

Open-source contributions
MindSpore / MindFormers

Research output
7 publications · 3 patent families

Core Areas

Parallelism: DP / TP / PP / EP · pipeline scheduling · recomputation

Planning: cost modelling · ILP · graph partitioning

Performance: profiling · trace analysis · bottleneck diagnosis

Technical Skills

Programming: Python, C++

Frameworks: PyTorch, Megatron-LM, MindSpore, MindFormers

Systems: Ascend CANN, HCCL

Tools: Linux, Git

Awards & Recognition

Best Poster Award — Euro-Par 2025
Letter of Appreciation
China Telecom · 2025
Golden Prize — Huawei, 2024

Languages

English · French · Mandarin · Cantonese

Online

ruiwen.wang
LinkedIn
Google Scholar

Experience

Ph.D. Researcher (CIFRE) — LLM Training Systems 2023 – 2026
Huawei Paris Research Center · Sorbonne Université & EURECOM · Paris

- Designed symbolic cost models and hybrid-parallel planners that jointly optimise parallelism, pipeline placement and recomputation under memory constraints; benchmarked against Megatron-LM.
- Integrated planners into MindSpore / MindFormers (open-source AI training stack; SAPP-PPB), enabling production use through configuration guidance, diagnostic traces and regression benchmarks across model families and cluster scales.

Research Apprentice → Research Engineer — AI Infrastructure 2021 – 2023
Huawei Paris Research Center · Distributed & Parallel Computing Lab · Paris

- Profiled and tuned large-scale LLM training and inference on production Ascend 910 NPU clusters, using execution-trace analysis to diagnose compute, memory and communication bottlenecks.

Deployments & Impact

- **Scale:** Ascend 910 cluster evaluations up to 16,384 NPUs. Model coverage across projects: dense, MoE and multimodal models, including models with over 1T parameters (DeepSeek, Llama, Mistral / Mixtral, Qwen).
- **Production training:** supported China Telecom's DeepSeek-MoE deployment (Letter of Appreciation) and TeleChat3-MoE training.

Publications

First-author publications

- **Concord** — NeurIPS 2026. Model edits trigger repeated planning work. Dependency and signature checks reuse valid layer-cost evaluations across model revisions, reducing re-planning time without changing the underlying search procedure.
- **PRISM** — SCA/HPC Asia 2026. Memory feasibility depends on parallelism, scheduling and recomputation. Symbolic memory and communication models evaluate these choices without per-plan profiling, enabling pre-execution peak-memory prediction.
- **BMPipe** — IEEE CLUSTER 2025. Pipeline stages waste compute time and leave memory unused. Joint symbolic modelling and ILP optimise placement, chunking and recomputation under memory limits, improving throughput at production scale.
- **H^2O** — Euro-Par 2025, Best Poster Award. Joint planning is combinatorial. A cost-guided hierarchy couples parallelism and micro-batching with pipeline placement and recomputation, improving throughput over automated and expert-tuned plans.
- **ManuMatic** — IFIP NPC 2025. Cost models can miss runtime overlap and new kernels. A few user-specified operator shardings steer recursive partitioning and graph-consistent completion, improving throughput over the automatic baseline.

Co-authored publications

- MLLM Pipeline Bubble Modeling — SCA/HPC Asia 2026 (poster).
- TeleChat3-MoE — arXiv 2025 (technical report).

Patents

- Execution-plan generation — WO 2025/020165 A1 (PCT), Huawei; published 2025.
- Large-model load balancing — Huawei; pending publication.
- Multidimensional parallelism configuration — Huawei; pending publication.

Education

Ph.D. Computer Science (CIFRE) — Sorbonne Université 2026 (defended)
M.Sc. Computer Science — Sorbonne Université 2019 – 2022
B.Sc. Computer Science — Université Paris-Saclay 2016 – 2019